

Retail Sales Data Dashboard

Minor Project — Data Analytics

1. Introduction

Retail organizations generate large volumes of transactional data. This project transforms retail transaction data into useful business information through cleaning, transformation, KPI development, exploratory analysis, and an interactive Power BI dashboard.

2. Business Objectives

Understand sales trends; identify top products and categories; compare regional performance; evaluate customer segments; monitor sales, profit, quantity, orders, discounts and profit margin; and identify loss-making products or sub-categories.

3. Dataset

The assignment specifies the Kaggle Superstore Dataset by Vivek Chowdhury. It contains transaction-level order, customer, geography, product, sales, quantity, discount and profit fields.

4. Methodology

Data collection → data-type validation → duplicate checking → missing-value handling → date feature creation → KPI calculation → aggregation by business dimensions → Power BI dashboard → business validation → recommendations.

5. Key Results

Standard dataset totals: 9,994 transactions; total sales \$2,297,200.86; total profit \$286,397.02; total quantity 37,873; profit margin approximately 12.47%.

6. Regional Results

Central: \$501,239.89 sales / \$39,706.36 profit. East: \$678,781.24 / \$91,522.78. South: \$391,721.91 / \$46,749.43. West: \$725,457.82 / \$108,418.45.

7. Category Results

Furniture: \$741,999.80 sales / \$18,451.27 profit. Office Supplies: \$719,047.03 / \$122,490.80. Technology: \$836,154.03 / \$145,454.95.

8. Findings

Technology is the strongest category by sales and profit in the standard dataset. West leads the regions in sales and profit. Furniture has substantial sales but a comparatively smaller profit contribution. Some sub-categories are loss-making, so discount and pricing analysis is important.

9. Recommendations

Protect and expand high-margin products; investigate loss-making sub-categories; review discount levels; use regional performance for resource allocation; monitor monthly sales and profit; and tailor promotions by customer segment.

10. Dashboard

The recommended Power BI dashboard contains KPI cards, monthly sales trend, regional comparison, category/sub-category analysis, top/bottom product analysis, customer-segment analysis, and interactive slicers.

11. Limitations

This is an educational Superstore dataset rather than current company data. Correlation between discount and profit does not prove causation. The project is primarily descriptive and diagnostic analytics.

12. Conclusion

The project demonstrates an end-to-end analytics workflow from raw transactional data to business recommendations and management reporting.